certainly, is yes. After all, space is space—regardless of whether it is tightly curled up into a Calabi-Yau shape or is unfurled into the grand expanse of the universe we perceive on a clear, starry night. In fact, we have seen earlier that the familiar spatial dimensions might themselves actually be curled up into the form of a giant shape that curves back on itself, way on the other side of the universe, and that therefore even the distinction between which dimensions are curled up and which are unfurled is somewhat artificial. Although our and Witten's analyses did rely on special mathematical features of Calabi-Yau shapes, the result—that the fabric of space can tear—is certainly of wider applicability.

Second, could such a topology-changing tear happen today or tomorrow? Could it have happened in the past? Yes. Experimental measurements of elementary particle masses show their values to be quite stable over time. But if we head back to the earliest epochs following the big bang, even non-string-based theories invoke important periods during which elementary particle masses do change over time. These periods, from a string-theoretic perspective, could certainly have involved the topology-changing tears discussed in this chapter. Closer to the present, the observed stability of elementary particle masses implies that if the universe is currently undergoing a topology-changing spatial tear, it must be doing it exceedingly slowly—so slowly that its effect on elementary particle masses is smaller than our present experimental sensitivity. Remarkably, so long as this condition is met, the universe could currently be in the midst of a spatial rupture. If it were occurring slowly enough, we would not even know it was happening. This is one of those rare instances in physics in which the lack of a striking observable phenomenon is cause for great excitement. The absence of an observable calamitous consequence from such an exotic geometrical evolution is testament to how far beyond Einstein's expectations string theory has gone.